



Molecular dynamics simulation of desalination process using polyoxometalate ionic liquid confined in carbon nanotubes

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ABSTRACT

In this work, the effect of positioning of [Keggin][emim]₃ ionic liquid (IL) into carbon nanotubes (CNTs) with different concentrations has been examined in the desalination process at different external pressures using molecular dynamics (MD) simulations. Another membrane was also modelled composed of four similar CNTs with almost equal space from each other. Using of only Keggin anions into the charged CNT was also studied in the desalination process. To study the mechanism of the water and ions fluxes, the potential of mean force, average number of hydrogen bonds, and self-diffusion coefficient have been also computed and discussed. The results showed that the placing of [Keggin][emim]₃ IL into a CNT decreases the water flux ($\frac{N_{water}}{ns} = 74at150MPa$) relative to the empty CNT ($\frac{N_{water}}{ns} = 834$). However, the four CNTs configuration shows increase in the water flux ($\frac{N_{water}}{ns} = 183at150MPa$) than the one CNT system. The placing of one polyoxometalate IL into the CNT increases the salt rejection 10 to 20 times than the empty-CNT system at low and high pressures, respectively. Moreover, positioning of the two ILs results in the complete salt rejection at all pressures. These results indicate that the use of polyoxometalate ILs into the CNT efficiently increases the salt rejection. The self-diffusion of confined water molecules in the single IL-CNT composite ($D = 0.73 \times 10^{-9} m^2/s$) is also smaller than the empty CNT at 150 MPa ($D = 1.89 \times 10^{-9} m^2/s$). The results also indicated that using of the Keggin anions into the charged CNT leads to the more water flux ($\frac{N_{water}}{ns} = 250at150MPa$) and complete salt rejection relative to the fixing of [Keggin][emim]₃ IL into the uncharged CNT. The effects of CNT charge distributions, temperature, and CNT chirality have been also examined on the desalination process.

1. Introduction

Developments in nanotechnology have created new paths to desalinate water with low cost and high efficiency [1,2]. Carbon nanotubes (CNTs) are excellent candidates for ion selection and water desalination because of their simple surface chemistry and nanoscale diameters. CNTs can transmit water rapidly and can be also simply functionalized [3–6]. CNTs also enable us to understand the mechanism of water transport in nanoscale dimensions which can be helpful for understanding the biological functions of cells [7,8].

Previous molecular dynamics (MD) simulations on desalination process using CNTs indicated important parameters which can affect the results. For instance, Corry [9] simulated desalination process using the CNTs with different sizes and found that the size and uniformity of CNTs are significant to get a desired result of salt rejection. In another work, Corry [10] examined the CNT functionalization and also the effect of charged CNTs in desalination process using MD simulation. He found that the electric charges at the entrance of CNT leads to more salt rejection but, the CNT functionalization results in the slower water flow.

Chan et al. [11] showed that zwitterion functional groups at the CNTs ends result in faster water transport and essentially rejects all ions using both experiment and MD simulations. Nasrabadi and Foroutan [12] simulated desalination process using the CNTs as nano-electrodes and found that the higher magnitude of applied charge leads to more ions separation from salty water. Khataee et al. [13] simulated desalination process using armchair silicon carbide (SiC) nanotubes by applying external electric field. They indicated that the (7, 7) nanotube can separate sodium ions but, the (8, 8) nanotube can separate chloride ions from solution selectively. Azamat et al. [14] also indicated high salt rejection rates using (5,5) boron nitride (BN) nanotube. Using MD simulations, Razmkhah et al. [15] indicated that strong electric field increases water transport through CNTs in desalination process but, magnetization of water molecules decreases water flow rate. Recently, Abbaspour et al. [16] studied the performance of CNT containing one and three fullerenes in desalination process using MD simulations and found that the presence of the fullerenes into the CNT leads to higher salt rejection rate. In a more recent work, Abbaspour et al. [17] examined hetero-junction nanomaterials using different doped CNTs in

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desalination process using MD simulations. They showed that the positioning of Ag-doped CNT and SiC nanotube at the first part of a heterojunction system decreases the water and ion fluxes but, the positioning of BN nanotube at the last part increases the ion flux.

Recently, the polyoxometalate anions such as Keggin has been used for dye absorption from aqueous solutions experimentally. For instance, Kahloul et al. [18] investigated the elimination of toluidine blue dye using ultra-filtration membrane using keggin polyoxometalates as complexing agents. They suggested using of polyoxometalates in place of polyelectrolytes and surfactants in the dye absorption process. More recently, Palapa et al. [19] intercalated the Keggin polyoxometalate anion with NiAl-layered double hydroxide and successfully used the synthesized material as an adsorbent for the removal of malachite green (MG) dye from aqueous solutions.

In this study, the performance of positioning of polyoxometalate ionic liquid ([Keggin][emim]₃ IL) into CNTs with different concentrations has been examined in the desalination process at different external pressures (150, 200, and 250 MPa) using MD simulations. Using of only Keggin anion with charged CNT has been also examined in the desalination process. In order to study the mechanism of the water and ions fluxes, the potential of mean force, average number of hydrogen bonds, self-diffusion coefficient, and angle distribution function have been calculated and discussed in this work.

2. Simulation details

The model has been built from two graphene surfaces with the size of 44.50 Å × 50.50 Å comprising a membrane spanned by a (25,0) CNT with the length of 39 Å. A membrane was also modelled by four similar CNTs which almost spaced equally from each other (please refer to Fig. S1 in the supporting information). One and two [Keggin][emim]₃ IL were also located into the CNT in different simulations. The structures of [Keggin] anion and [emim] cation have been presented in Fig. S2 in the supporting information. The CNTs and both graphene surfaces were kept fixed during the MD simulations. Many previous researchers have also kept fixed the membrane during the simulation of desalination process [20–23]. The CNT with both graphene surfaces were positioned in the middle of simulation box with the dimensions of 50.5 Å × 44.5 Å × 130 Å (the length of the water source was 41.7 Å). The solution consisting of 3386 water molecules and 32 sodium and 32 chloride ions (almost according to the salt concentration in sea water, 33 g/L [24]) was used in the simulation box. To not escape the confined ILs from the CNT with high applied pressures, the ILs have been fixed into the CNT. However, the ILs were initially equilibrated into the CNT to obtain the most stable structures of the confined ILs into the CNT and then, they subjected to the desalination simulations. The [emim] cation was also modeled using the OPLS-AA potential [25]. The Keggin anion was modeled as a rigid particle and the all-atom force field parameters of the PW₁₂O₄₀³⁻ anion were taken from the work by Lopez et al. [26], which is based on the Amber force field and has been extensively applied for the polyoxometalate anions simulation [27,28]. Chaumont and Wipff also simulated Keggin anion using the same Amber force field in different solvents modeled using the OPLS force field [29]. We have also simulated the unfixed [Keggin] anion in charged CNT in which the electric charge was distributed uniformly between the atoms of the CNT (every atom had 0.2e electric charge which was the least charge needed not to escape the anion from the CNT at lowest applied pressure in this work, i. e. 150 MPa). The SPC/E model [30] was also used for water molecules. Previous studies on ability of different water models to calculate the thermodynamics, dynamics, and structural properties of confined water molecules into carbon nanostructure indicated that the results of some water models are almost similar but the SPC/E water model is the optimum choice [31,32]. The Lennard-Jones (LJ) model was used for the non bonded ion-ion, ion-water, and other interactions with the Lorentz-Berthelot combination rules [33–35]. The interactions of [Keggin][emim]₃ IL with the other particles were also modeled using the LJ

model. The LJ parameters for the carbon atoms were 0.2327 (kJ/mol) and 3.4 (Å) [36]. The different parameters of the force fields used in this work have been also presented in Tables S1 and S2 in the supporting information.

To apply a hydrostatic constant pressure to the systems (from left to right in the z direction), Zhu et al.'s [37,38] technique was used. In this way, a constant force is exerted to the oxygen atom of water molecules, according to the following relation:

$$P(Pa) = \frac{nF(N)}{A(m^2)} \quad (1)$$

where n is the number of water molecules, F is the value of induced constant force, P is the applied pressure, and A is the area of the box. This technique has been used in many simulations of pressure-driven flow [39–47] including my previous works [16,17].

All of the MD simulations have been carried out in the NVT ensemble using the Nosé–Hoover thermostat at 300 K with the modified DL-POLY software [48]. The Ewald summation method was used for the electrostatic interactions. The cutoff distance of 12 Å was used. The periodic boundary conditions have been also used in three directions. After the solution reached the equilibrium, the MD simulations have been performed for 5 ns with the time step of 1 fs. The empty CNTs (CNTs without ILs) systems have been simulated for 3 ns.

In the first step, a wall was positioned at the entrance of the CNT to hinder the passing of solution (at the left side of the membrane) through the tube and reach the solution to the equilibrium for 2 ns (Initially, the right side of the membrane was empty). Then, the wall was removed and the none-equilibrium MD simulations have been performed by applying the external pressure to the water molecules for 5 ns with the time step of 1 fs. The empty-CNT (CNT without ILs) systems have been simulated for 3 ns. The simulations have been repeated three times and the average values have been presented with the standard deviations in Figs. 1 and 3.

3. Results and discussion

3.1. Water and ion fluxes

The water flux (the number of water molecules transmitted from the CNT per nanosecond) has been given for the Empty-CNT (the empty CNT without any IL), 1IL-1CNT (one IL in one CNT), 2ILs-1CNT (two ILs in one CNT), 4ILs-4CNTs (one IL in every CNT of the four-CNTs configuration), and 8ILs-4CNTs (two ILs in every CNT of the four-CNTs configuration) systems at different pressures of 150, 200, and 250 MPa at 300 K in Fig. 1. For example, the snapshot of the final configuration of 8ILs-4CNTs system has been presented after 5 ns of simulation

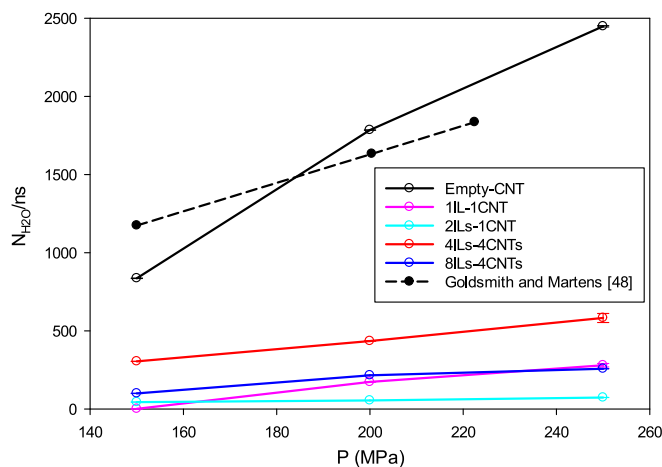


Fig. 1. The water flux in the different IL-CNT systems at different pressures at 300 K.

times in Fig. 2.

According to Fig. 1, the simulated water flux using the empty-CNT system is almost in good agreement with the simulated results of Goldsmith and Martens [48] on desalination process using (14,14) CNT (The difference between the two results can be due to the different force fields and the method of desalination simulation). It is also shown that the water flux increases by increasing the external pressure almost linearly for all IL-CNT systems. According to Fig. 1, the positioning of [Keggin][emim]₃ IL in the CNTs decreases the water flux. The confined polyoxometalate IL hinders the water transport within the CNTs which leads to the smaller water flux than the empty-CNT system. Presence of more polyoxometalate ILs in the CNTs also decreases the water flux more. It is also shown that the 4CNTs configurations significantly increases the water flux than the 1CNT systems. The water flux of 4ILs-4CNTs is almost two times more than that of the 1IL-1CNT system at lower pressure. The water flux of 8ILs-4CNT system is 4 and 6 times more than that of the 2ILs-1CNT system at low and high pressures, respectively.

The ion flux (the number of Na⁺ plus Cl⁻ ions transmitted from CNT per nanosecond) has been also presented for the Empty-CNT, 1IL-1CNT, 2ILs-1CNT, 4ILs-4CNTs, and 8ILs-4CNTs systems at different pressures at 300 K in Fig. 3. According to this figure, the simulated ion flux using the empty-CNT system is in good agreement with the simulated results of Goldsmith and Martens [48] on desalination process using (14,14) CNT at smaller pressure. It is also shown that the positioning of one polyoxometalate IL in the CNT (1IL-1CNT system) increases the salt rejection 10 to 20 times than the empty-CNT system at low and high pressures, respectively. Moreover, positioning of the two ILs results in the complete salt rejection at all pressures. These results indicate that the use of polyoxometalate ILs into the CNT efficiently increases the salt rejection. It is also shown that although using of the 4CNTs

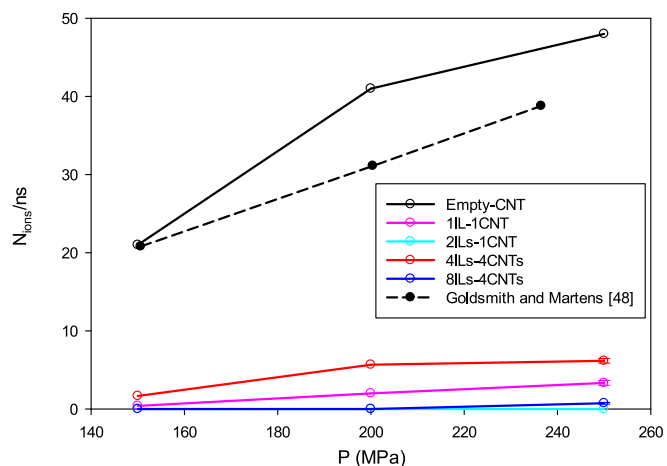


Fig. 3. The ion flux in the different IL-CNT systems at different pressures at 300 K.

configurations increases the ion flux than the 1CNT systems, the 8ILs-4CNTs system represents almost the complete salt rejection. Therefore, using of eight [Keggin][emim]₃ ILs in the 4CNTs configuration leads to the acceptable water flux and nearly complete salt rejection than the other IL-CNT systems.

3.2. Potential of mean force

In order to examine the effect of confined polyoxometalate ILs into the CNT on the water and ion fluxes in desalination process, the potential of mean force of water and ion penetrations (PMF(z)) have been

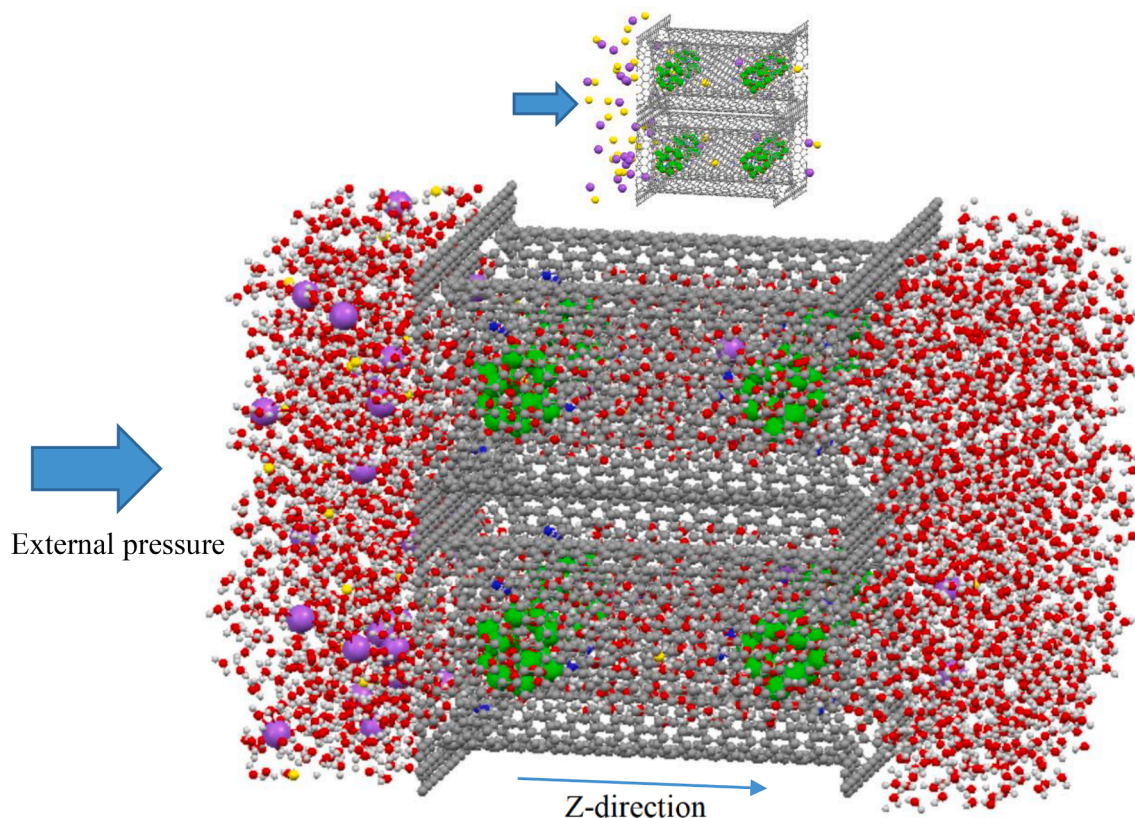


Fig. 2. The snapshot of the final configurations of the 8ILs- 4CNTs system after 5 ns of the simulation times at $P = 250$ MPa and 300 K. The C atoms are in gray, O atoms are in red, H atoms are in white, Na⁺ ions are in violet, and Cl⁻ ions are in yellow, W atoms are in green, P atom is in dark green, and N atoms are in blue. The water molecules have been deleted in the above snapshot.

computed at 150 MPa and presented in Fig. 4. The PMF (z) indicates the potential barrier energy needed to water and ions to penetrate and cross the CNT [49] and can be calculated from the following relation:

$$PMF(z) = -k_b T \ln \left(\int \frac{\rho(z, r)}{\rho} dr \right) = -k_b T \ln \left(\frac{\rho(z)}{\rho} \right) \quad (2)$$

where k_b is Boltzmann constant, T is temperature, and ρ is the number density.

According to Fig. 4, the water and ions undergo a potential barrier energy before entrance to the CNT. The water and ions experience highest potential barrier energy at the entrance to the CNT with two confined ILs which results in the smaller water and ion fluxes. The water and ions confined into the CNT also experience higher potential barrier energy at the positions of the [Keggin][emim]₃ IL. This barrier is also higher for the CNT with more confined polyoxometallate ILs. The barrier energy achieves to infinity range for the ions at the positions of the polyoxometallate ILs (Fig. 4b).

3.3. Average number of hydrogen bonds

The average number of hydrogen bonds per water molecule ($\langle \text{HB} \rangle$) was also calculated for the IL-CNT systems for the first 1 ns of simulation time at 150 MPa and presented in Fig. 5. The criteria were the position of the first O-O RDF peak as the HB cutoff distance (2.88 Å) and the HB angle of 150° [17]. As Fig. 5 shows, the $\langle \text{HB} \rangle$ initially decreases for the systems because the water molecules transit from a bulk environment to the nano space of the CNT and thus the $\langle \text{HB} \rangle$ decreases. It is also shown at the initial simulation times which related to the first entrance of the water molecules into the CNT, the more

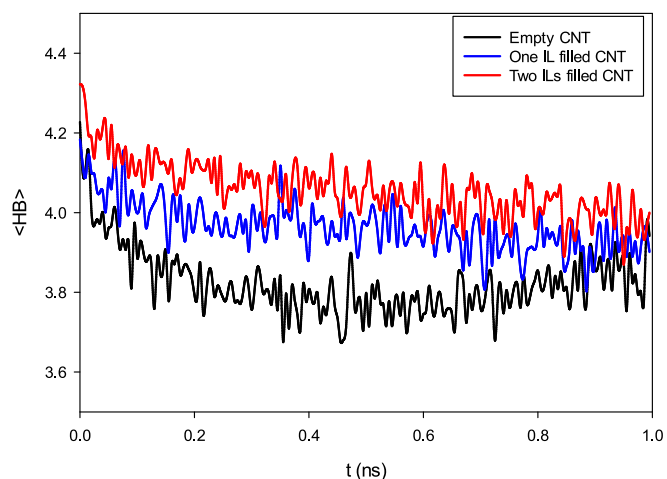


Fig. 5. The average number of hydrogen bonds per water molecule ($\langle \text{HB} \rangle$) for the different IL-CNT systems at 150 MPa at the first 1 ns of simulation times.

polyoxometallate ILs results in the more $\langle \text{HB} \rangle$. This result is in agreement with the potential barrier energy result in Fig. 4. In the other words, the higher potential barrier energy in addition to the higher $\langle \text{HB} \rangle$ lead to the less water and ion fluxes in the IL-CNT systems with more ILs.

3.4. Self-diffusion coefficient

In order to investigate the dynamics of the confined water molecules into the CNT in different systems, the CNT and its containing in the different systems were separated from the remaining of the systems and simulated for 3 ns in the NVT ensemble with equal number of water and ions (the excess molecules were deleted). The self-diffusion coefficients of confined water molecules were computed from the slopes of mean square displacement (MSD) plots using the following equation [50] and presented in Fig. 6:

$$D = \lim_{t \rightarrow \infty} \frac{1}{6tN} \sum_{i=1, N} \langle |r_i(t) - r_i(0)|^2 \rangle \quad (3)$$

Where N is the number of particles and the positions of the particles at time $t = 0$ and t are shown by $r_i(0)$ and $r_i(t)$, respectively. As Fig. 6 shows, the self-diffusion coefficient of confined water molecules decreases by increasing the confined polyoxometallate ILs into the CNT which is due to the increasing water-IL interactions. Thus, in addition to

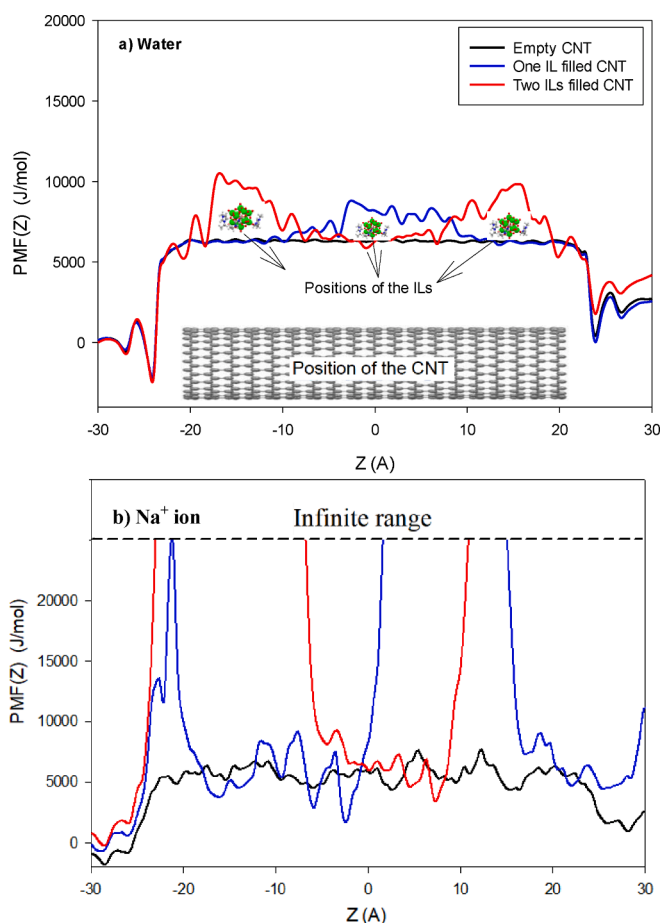


Fig. 4. The potential of mean force of a) water and b) Na^+ ion penetrations for the IL-CNT systems at 150 MPa and 300 K.

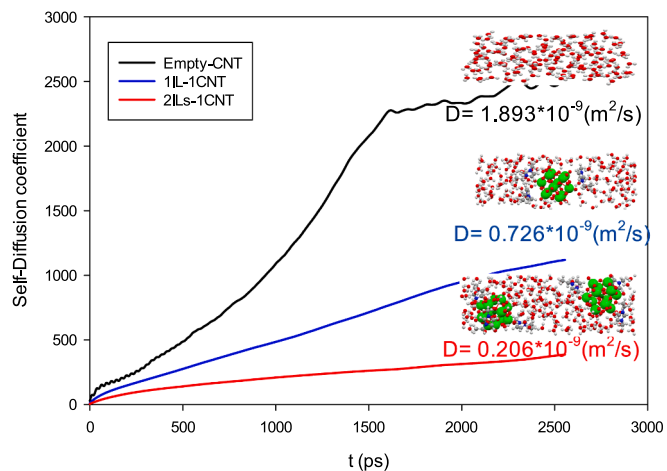


Fig. 6. The self-diffusion coefficient of confined water molecules into the CNT at different systems.

the more < HB > and higher potential barrier energy, the other reason of the smaller water flux in the CNTs with more confined polyoxometallate ILs is the stronger water-IL interaction which results in the smaller self-diffusion of the confined water molecules. The decrease in the accessible volume to the confined water molecules is another reason of the smaller self-diffusion of the confined molecules by increasing the confined ILs.

3.5. The [Keggin] anion into charged CNT

The external pressures used in the real desalination apparatus is much less than those of these simulations (less than 10 MPa [51]) and we should not worry about the exit of the ILs from the nanostructures. However, one of the solutions is proposed here: using of the Keggin anions into positively charged CNT. We have also used charged CNT in the MD simulation of desalination process using fullerene-filled CNTs [16] (not to escape the fullerene from the CNT). Goldsmith and Martens [48] also examined charge distributions on the CNT in desalination process using MD simulations. To examine this idea, One [Keggin] anion into the CNT with 0.2e positive charge on every carbon atom was simulated at external pressure of 150 MPa (Fig. 7) for 1 ns and the related water and (chloride) ion fluxes have been presented in Fig. 8 (any water or ions could not transfer after almost 1 ns because of the blockage of the CNT by the ions). As Fig. 8 shows, the [Keggin]-charged CNT system shows higher water flux than the 1IL-1CNT system up to five times. One of the reason of this result is due to the lack of [emim] cations into the charged CNT which makes more space to the confined water molecules to transmit through the CNT. However, it is shown that the Keggin anion into charged CNT leads to the complete salt rejection. This is due to the fact that the CNT is positively charged and thus, repel Na^+ ions and attract Cl^- ions and collect them into the CNT (Fig. 7). Thus, using of the Keggin anions into the charged CNT leads to the more water flux and complete salt rejection relative to the fixing of [Keggin] [emim]₃ IL into the uncharged CNT.

The potential of mean force of water penetration (PMF(z)) has been also computed for the [Keggin]-charged CNT system and compared with the 1IL-1CNT system at 150 MPa and 300 K in Fig. 9.

According to Fig. 9, the water molecules in the [Keggin]-charged CNT system experience lower potential barrier energy than the uncharged 1IL-1CNT system before entrance to the CNT which results in the more water flux in the charged CNT system. Therefore, in addition to the more internal space of the [Keggin]-charged CNT system than the 1IL-1CNT system, the lower potential of mean force of water penetration is another reason of the more water flux in the charged CNT system. The confined water molecules also experience higher potential barrier energy at the positions of the polyoxometallate IL and the Keggin anion.

To examine the charge distribution on CNT atoms, we have distributed the positive charges (0.2e on every carbon atom) only on the first, center, and the last (one-third) part of the CNT atoms and compare the corresponding simulation results of water flux in Fig. 10 (In all of the

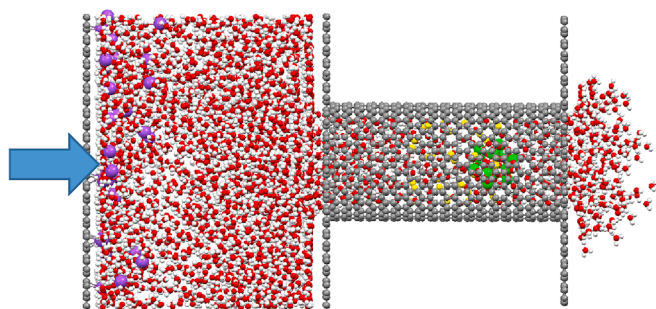


Fig. 7. The snapshot of the configuration of the [Keggin]-charged CNT system at $P = 150$ MPa and 300 K. The C atoms are in gray, O atoms are in red, H atoms are in white, Na^+ ions are in violet, and Cl^- ions are in yellow, W atoms are in green, and P atom is in dark green.

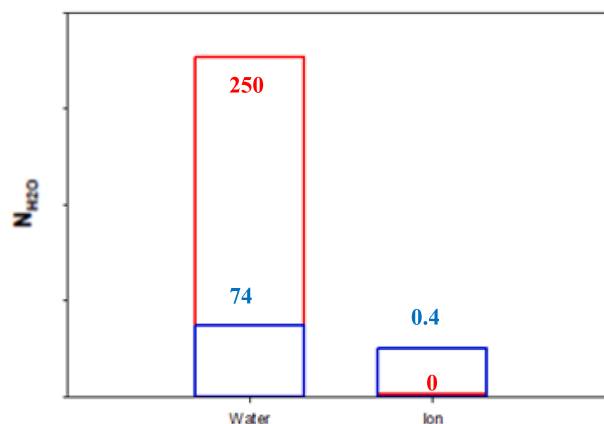


Fig. 8. Comparison of the number of water and ions between the [Keggin]-charged CNT (red bars) and 1IL-1CNT (blue bars) systems get out from the CNT after 1 ns at 150 MPa and 300 K.

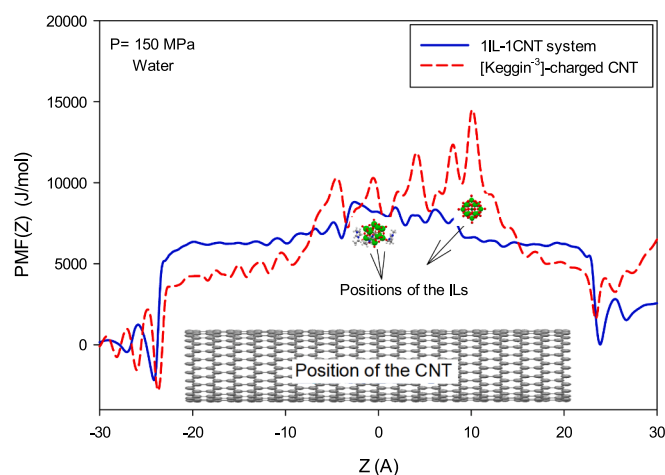


Fig. 9. Comparison of potential of mean force of water penetration between the [Keggin]-charged CNT system (red dashed line) and the 1IL-1CNT system (blue bars) at 150 MPa and 300 K.

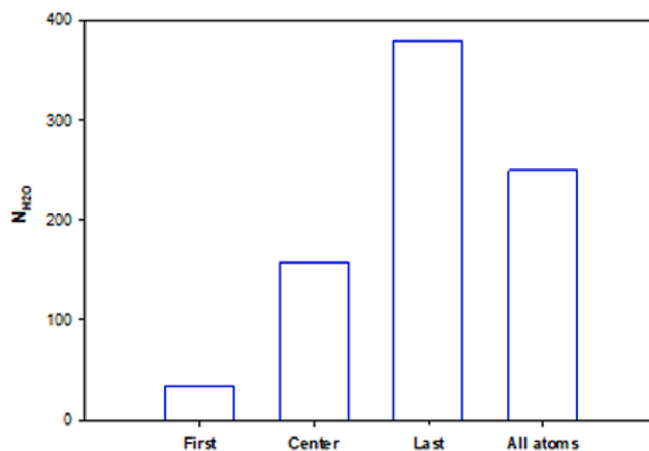


Fig. 10. Effect of the charge distribution on CNT atoms on the number of water molecules get out from the CNT after 1 ns at 150 MPa.

charge distribution forms, the ion rejection was complete). As Fig. 10 shows, positioning the positive charges on the last part of the CNT leads to the more water flux than the other configurations.

3.6. Temperature effect

To examine the temperature effect on the results, we have presented the water and ion flux for 1CNT-1IL system at 300, 320, 350, and 370 K in Fig. 11. As this figure shows, the water and ion flux increases by increasing the temperature which is due to the increasing of the kinetic energy of the particles. Although the water flux increases, the ion reject rate decreases. This is due to the fact that the intermolecular interaction decrease by increasing the temperature and the efficiency of the membrane decreases.

3.7. CNT chirality effect

To examine the effect of CNT chirality on the results, we have presented the water and ion flux for 1CNT-1IL system using the zigzag (25,0) and armchair (15,15) CNTs with almost similar diameters (the diameters of zigzag and armchair CNTs are 19.60 Å and 20.39 Å, respectively) at 150 MPa and 300 K in Fig. 12. As this figure shows, the water and ion fluxes are less in the zigzag CNT than the armchair one. Part of this result is due to (a bit) smaller diameter of the zigzag CNT than the armchair one. The other reason is due to the fact that the confined molecules experience greater interactions in the zigzag CNT than the armchair CNT and this leads to the more ion rejection. This result is also in agreement with the results of Akbarzadeh et al. [52] on

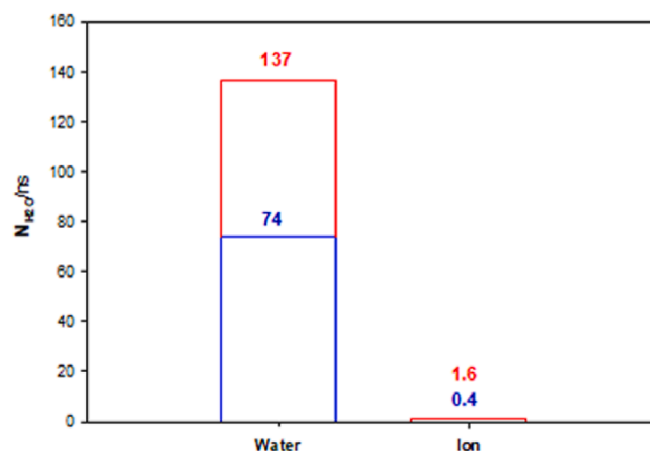


Fig. 12. Comparison of water and ion fluxes between in the 1IL-1CNT system using zigzag (25,0) CNT (blue bars) and armchair (15,15) CNT 1IL-1CNT (red bars) at 150 MPa and 300 K.

MD simulation of confined molecules ionic liquid molecules in different zigzag and armchair CNTs. They concluded that the confined molecules have greater interactions with the zigzag CNT than the armchair one.

4. Concluding remarks

In this work, the effect of positioning of [Keggin][emim]₃ IL into CNT with different concentrations (1IL-1CNT and 2ILs-1CNT) has been examined in the desalination process at different external pressures (150, 200, and 250 MPa) using MD simulations. Another membrane was also modelled composed of four similar CNTs with almost equal space from each other. Using of only Keggin anions into the charged CNT was also studied in the desalination process. To study the mechanism of the water and ions fluxes, the potential of mean force, average number of hydrogen bonds, and self-diffusion coefficient have been computed and the following important results have been concluded:

- 1) The placing of [Keggin][emim]₃ IL into the CNTs decreases the water flux. Presence of more ILs in the CNTs also decreases the water flux more.
- 2) The 4CNTs configurations significantly increases the water flux than the 1CNT systems. The water flux of 4ILs-4CNTs is almost two times more than that of the 1IL-1CNT system at lower pressure. The water flux of 8ILs-4CNT system is 4 and 6 times more than that of the 2ILs-1CNT system at low and high pressures, respectively.
- 3) The placing of one polyoxometallate IL into the CNT increases the salt rejection 10 to 20 times than the empty-CNT system at low and high pressures, respectively. Moreover, positioning of the two ILs results in the complete salt rejection at all pressures. These results indicate that the use of polyoxometallate ILs into the CNT efficiently increases the salt rejection.
- 4) Using of eight [Keggin][emim]₃ ILs in the 4CNTs configuration leads to the acceptable water flux and nearly complete salt rejection than the other IL-CNT systems.
- 5) The higher potential barrier energy in addition to the higher $\langle \text{HB} \rangle$ lead to the less water and ion fluxes in the IL-CNT systems with more confined ILs.
- 6) The self-diffusion coefficient of confined water molecules decreases by increasing the confined polyoxometallate ILs into the CNT which is due to the increasing water-IL interactions. Thus, in addition to the more $\langle \text{HB} \rangle$ and higher potential barrier energy, the other reason of the smaller water flux in the CNTs with more confined polyoxometallate ILs is the stronger water-IL

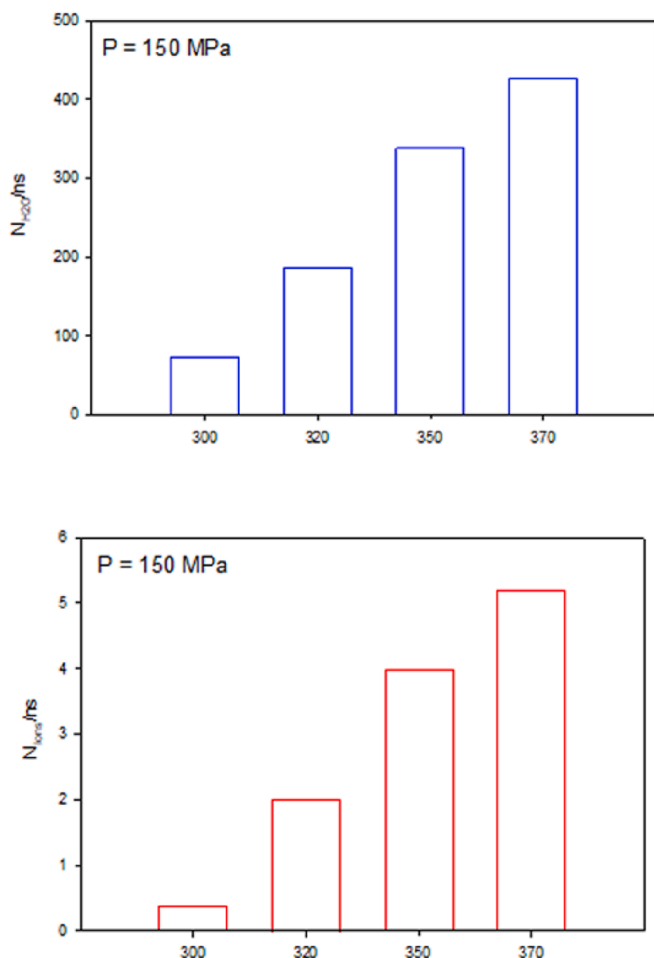


Fig. 11. Comparison of water (blue) and ion (red) fluxes in the 1IL-1CNT system at 150 MPa at different temperatures.

interactions which results in the smaller self-diffusion of the confined water molecules.

- 7) Using of the Keggin anions into the charged CNT leads to the more water flux and complete salt rejection relative to the fixing of [Keggin][emim]₃ IL into the uncharged CNT. The lower potential of mean force of water penetration is a reason of the more water flux in the charged CNT system.
- 8) positioning the positive charges on the last part of the CNT leads to the more water flux than the other configurations.
- 9) The water and ion flux increases by increasing the temperature. Although the water flux increases, the ion reject rate decreases.
- 10) The water and ion fluxes are less in the zigzag CNT than the armchair one.

CRediT authorship contribution statement

Mohsen Abbaspour: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - original draft, Review and editing, Visualization, Supervision, Project administration. **Majid Namayandeh Jorabchi:** Software, Resources, Review and Editing, Validation. **Hamed Akbarzadeh:** Review and Editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.molliq.2023.122310>.

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